

# Micro-services Labs: Polymove

Fill the forms with your information once completed:

<https://forms.gle/Qyn6vWMoxZ1PkpRF9>

In this first lab; we will implement a micro-service architecture called Polymove.

Polymove is a platform that connects university students with international internship opportunities. The architecture is composed of multiple independent services, each with its own responsibility and database.

Polymove will drive you through the implementation of services and pattern related to micro-services architecture.

For all this lab, you can write the service with the language of your choice between one of the following: TypeScript, Rust, Golang or Python.

## Let's start with our first service: Polytech

### Objective

The first service is called Polytech and is basically where you study. In our case, we will focus on the listing of students and their internship registration.

### Composition

This service exposes a REST API to manage students. Students are saved in a Postgres database.

### Data Model

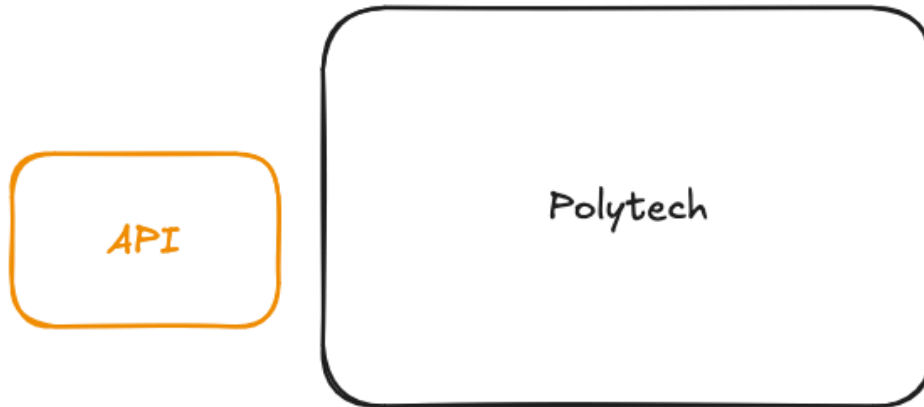
A student is associated to the following: id, firstname, name and a domain (IT, life science, ...)

### Endpoints to Implement

| Method | Endpoint                                    | Description                           |
|--------|---------------------------------------------|---------------------------------------|
| POST   | <code>/student</code>                       | Register a new student                |
| GET    | <code>/student/:id</code>                   | Retrieve a student by ID              |
| GET    | <code>/student?domain=&lt;domain&gt;</code> | Retrieve all students within a domain |

|        |              |                               |
|--------|--------------|-------------------------------|
| PUT    | /student/:id | Update the field of a student |
| DELETE | /student/:id | Delete a student              |

Register students in the university



POST /student  
ID  
Firstname  
Name  
Domain

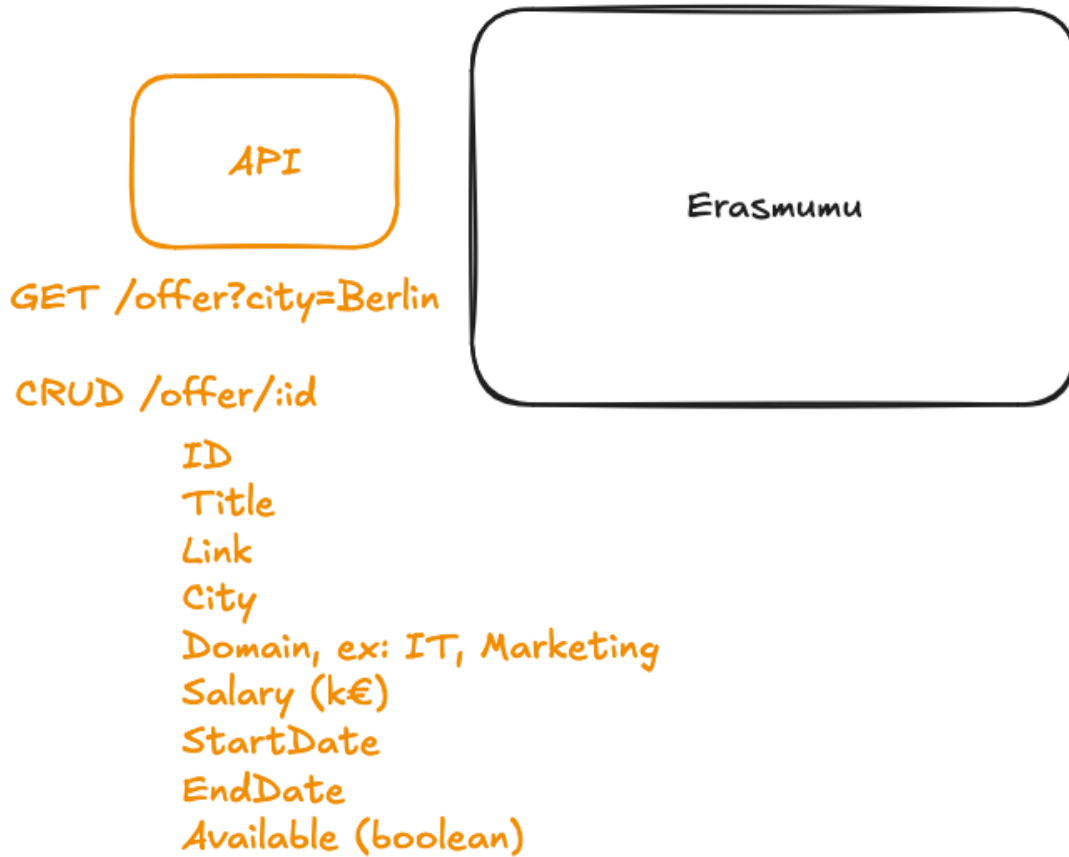
GET /student?domain=devops  
ID  
Forname  
Name  
Domain

## Tests

Ensure your endpoints works as expected. Once done, ensure

## Implement the Internship offer service: Erasmumu

In this step, you will implement the Erasmumu service. Erasmumu helps students to find internship offers abroad, by exposing offers. Erasmumu is connected to a MongoDB Database.



Erasmumu is the online offer services for student

Allow Company to post offers  
Allow student to retrieve offers

### Data Model

An offer is composed of the following

- ID
- Title
- Link
- City
- Domain
- Salary
- StartDate
- EndDate
- Available

## Endpoints to Implement

| Method | Endpoint                                                                                                      | Description                                                                                       |
|--------|---------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| POST   | <code>/offer</code>                                                                                           | Create a new offer                                                                                |
| PUT    | <code>/offer/:id</code>                                                                                       | Update an existing offer                                                                          |
| GET    | <code>/offer/:id</code><br><code>/offer?domain=&lt;domain&gt;</code><br><code>/offer?city=&lt;city&gt;</code> | Retrieve an offer by ID<br>Retrieve offers filtered by domain<br>Retrieve offers filtered by city |
| DELETE | <code>/offer/:id</code>                                                                                       | Remove an existing offer                                                                          |

## Rules

- An offer **must not** be returned if `available` is `false`
- Proper error codes must be implemented

## Goals

- Populate the services with different offers: put different domains and different cities
- Ensure your routes works as expected.

## Inter-services communication: Internship registration

We have the offers and the student, now we need the registration system. To be close from reality, you need to pass the offers to Polytech. They will ensure the offer match your domain.

### Objective

Implement the registration flow that connects both services. A student registers for an internship through Polytech, which validates the request by communicating with Erasmumu.

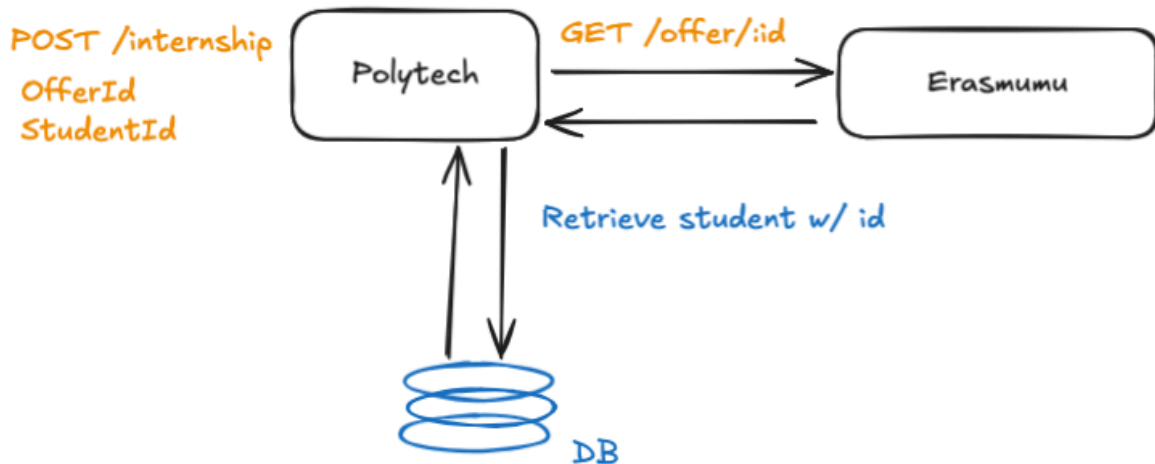
## New Endpoints for Polytech Service

| Method | Endpoint                 | Description                          |
|--------|--------------------------|--------------------------------------|
| POST   | <code>/internship</code> | Register a student for an internship |

GET      /internship/:id      Retrieve the status of a registration

Here is the simplified flow of the POST /internship request

1. University retrieve the offer to Erasmumu
2. University retrieve the student in DB
3. University ensure: offer domain = student domain
4. University reply with the status



#### DataModel: Internship registration

- Id
- StudentId
- offerId
- Status (approved or rejected)
- message

### Example of registration approved

STUDENT  
ID: 003b-7d8a  
Firstname: Leo  
Name: Bronsin  
Domain: DevOps

OFFER  
ID: ab37-45kl  
Title: Enhance CI/CD pipelines for dev teams  
...  
Domain: DevOps

ID: <uuid>  
Status: Approved  
Message: Student succesfully registered

### Example of registration rejected :

STUDENT  
ID: 003b-7d8a  
Firstname: Leo  
Name: Bronsin  
Domain: DevOps

OFFER  
ID: 8a37-45kl  
Title: Snorkling with the Orca in Polar Ocean  
...  
Domain: Life Science

ID: <uuid>  
Status: Rejected  
Message: Offer domain doesn't match

### Goal

- Implement the new routes and the business logic to save the internship.
- Make sure to handle edge cases

### Deliverable

Once completed, commit your work to a branch named:

lab-synchrone-com-polytech-erasmumu

Fill the forms with your information once completed:

<https://forms.gle/Qyn6vWMoxZ1PkpRF9>

Be sure to put your repository in public or invite me with my [GridexX](#)